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SUBJECT- SPEECH AND NATURAL LANGUAGE PROCESSING

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ASSIGNMENT-1

1) Explain the role of Tokenisation in Natural Language Processing.

→ Tokenization is the process of breaking a text into smaller units called tokens, such as words, sub-words or sentences. It is one of the first and most crucial preprocessing steps in Natural Language Processing.

Role of tokenization-

1) Text Segmentation - Tokenization divides raw text into meaningful units (tokens) like:

- word tokens: "NLP is powerful" → [NLP, is, powerful]
- sentence tokens.
- subwords tokens used in modern NLP models]

2) Foundation for further processing - Most NLP tasks such as stemming, lemmatization, POS tagging, parsing, and sentiment analysis, require tokenized input, without tokenization machines cannot interpret text structure.

3) Vocabulary creation -

Tokenization helps in building a vocabulary of unique words or subwords which is essential for feature extraction technique like Bag of words, and TF-IDF

(A) Improves computational efficiency - By converting large text into tokens, NLP systems can process data faster and more efficiently.

5) Handling Language Variations -
Tokenization manages punctuation, special characters, contractions and different writing styles.

6) Support Machine Learning Models: ~~Model~~ ML and DL models cannot process raw text. Tokenization converts text into structured formats suitable for numerical encoding.

Types of Tokenization

- word tokenization
- Sentence "
- Subword tokenization (BPE, word piece)

② Analyze the differences between syntax, semantics and pragmatics in NLP.

⇒ Natural language understanding involves interpreting human languages at different levels. Syntax, semantics and pragmatics are these core linguistic components.

1) Syntax - Syntax deals with the grammatical structure of sentences.

- focuses on word order and sentence formation
- Determines whether a sentence is grammatically correct.

Example - she is reading a book. ✓
Reading the book a is. X

2) Semantics - Semantics focuses on the meaning of words and sentences.

- Determines whether a sentence makes sense logically
- Deals with word meanings and sentence interpretation.

Example - "The cat chased the mouse." Meaning is clear.

"The mouse chased the cat." Different meaning

3) Pragmatics - Pragmatics deals with contextual meaning and speaker's intention.

- Depends on situation, context and real-world knowledge.
- Helps interpret implied meaning.

Example - 'Can you open the window?'
 → Pragmatically means a request not a question about ability

Comparison table -

Aspect	Syntax	Semantics	Pragmatics
Focus	Structure	Meaning	Context
Concern	Grammar	Interpretation	Intention
Dependency	Language rules	Word meaning	Situation

③ Illustrate the application of k-Nearest Neighbors (k-NN) algorithm in document classification.

⇒ The k-NN algorithm is a supervised instance-based learning algorithm commonly used for document classification in NLP.

Concept of k-NN - k-NN classifies a document based on the majority class of its k nearest documents in the training dataset.

Steps in Document Classification using k-NN

1) Text Preprocessing -

- Tokenization
- Stop word removal
- Stemming / Lemmatization

2) Feature Representation -

- Convert documents into numerical vectors using 'Bag of words', TF-IDF

3) Distance calculation

- Measure similarity using cosine similarity
- Euclidean Distance

4) Choosing k Nearest Neighbors

Identify the k most similar documents

5) Classification -

Assign the class that appears most frequently among neighbors.

Example - If a new document is similar as

to

- 3 "Sports" documents

- 2 "Politics" " "

→ It is classified as Sports

Advantages

- Simple and easy to implement

- No training phase required

- Effective for multi-class classification

Limitations -

- Computationally expensive for large datasets,

- Sensitive to choice of k .

- Requires efficient distance computation.